

GPU substructure for flashjet

kt / C-A jet substructure on the merge history

exclusive jets · soft-drop grooming · Lund coordinates

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Every claim below is closed against an independent reference and/or the defining paper. All plots regenerable: `plots/2026-07-13-substructure/` .

Summary — what this deck shows

Three substructure features (F1 exclusive-kt jets, F2 soft-drop/mMDT grooming, F3 Lund coordinates), all **pure-torch post-reads of the existing merge history** — no kernel changes, CPU/CUDA identical, negligible cost.

Validation ladder, each rung closed:

rung	reference	headline result
unit tests	independent NumPy tree-walks	85 passed (13 CUDA-only skipped)
paper closures	analytic predictions, toy shower	z_g on the $1/z$ curve; areas; β -ordering
real CMS QCD	stored FastJet branches (raw-to-raw)	p_T 1.000000 ; m_{SD} -0.004 GeV ; R_g 99.2% <0.01
real CMS ttbar ×2	stored FastJet branches (raw-to-raw)	dileptonic UL18 and fully-hadronic Run 3 close
full-event	ΔR -match to CMS's own AK8 jets	100% within 2%, median ΔR 0.0019
physics regimes	QCD vs b-jets vs boosted tops	m_W/m_t peaks, top blob in the Lund plane

flashjet reproduces CMS's FastJet reconstruction to NanoAOD storage precision. The $\sim 4\%$ m_{SD} tail is fully attributed (missing soft constituents + storage rounding + threshold flips — see outlier anatomy); relative agreement is $\sim 0.1\%$ at all masses.

Where this sits in flashjet

flashjet clusters padded `(B, N, 4)` torch tensors into jets on the GPU, staying on-device for ML training loops. Generalized-kt:

$$d_{ij} = \min(k_{t,i}^{2p}, k_{t,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad p = -1 \text{ (anti-} k_t), 0 \text{ (C/A), } +1 \text{ (} k_t)$$

The key object Alex's kernels already produce: the *merge history*. Every recombination step records four arrays — `hist_p1`, `hist_p2` (the two inputs), `hist_child` (the output id), `hist_d` (the distance at which they merged). That history *is* the full binary clustering tree.

Our contribution = read that tree. No new kernels, no changes to the clustering path. All three features are pure-torch post-reads of `(hist_p1, hist_p2, hist_child, hist_d)`, so they run unchanged on CPU or CUDA and add negligible cost.

What was already there vs. what we added

Pre-existing (Alex)

- `history.py`: `jet_idx_from_history`, `_resolve_parents`, `_decode`, `splitting_scales_from_history`, `Triton`, `_decode_triton`
- `reference.py`: single-event NumPy ground truth (`cluster_event`)
- `nn_reference.py`: nearest-neighbour reference
- `conftest.py`: `random_event` fixtures
- The kernels + the merge-history recording

The `_ref_*` walkers inside `test_substructure.py` are also ours — independent naive NumPy tree-walks used only to pin the fast implementations.

Added (this work) — in `history.py`

- **F1** `exclusive_jets_from_history`
- **F2** `groom_from_history`
- **F3** `lund_coordinates_from_history`
- helpers: `_resolve_roots`, `_jet_roots`, `_pseudojet_p4`, `_dense_number_roots`
- `api.py`: `ClusterOutput.{exclusive_jets, groomed_jets, mass_drop, lund_coordinates}` + a `mask` field
- `tests/test_substructure.py` (new)

The three features

	Feature	Function	Implements
F1	Exclusive jets (kt)	<code>exclusive_jets_from_history(...)</code>	kt algorithm — Fastjet manual [1111.6097], [0802.1189]
F2	Grooming (C/A)	<code>groom_from_history(...)</code>	Soft Drop [1402.2657] · mMDT [1307.0007] · mass-drop [0802.2470]
F3	Lund coordinates	<code>lund_coordinates_from_history(...)</code>	Primary Lund plane [1807.04758]

F1 — undo the last merges of the sequence to expose exactly `n_jets` (or a `d_cut`) subjects; reduces to the inclusive jets at the trivial cut.

F2 — walk each jet down the harder branch, dropping the softer prong until $z > z_{\text{cut}}(\Delta R/R)^\beta$ (soft-drop) — the $O(\log_2 n)$ declustering.

F3 — for every primary split emit $(z, \Delta R, k_t, \ln 1/\Delta R, \ln k_t, d)$; the d -channel ties **exactly** to `splitting_scales` .

How the toys were generated

(the "simulation" — none of it pre-existing; flashjet only clusters, it does not generate events)

The toy event generators (written in the plot scripts)

flashjet takes 4-vectors in and returns jets — it has **no event generator**. To exercise the features I wrote small, transparent toys *outside the repo*, in the plot scripts.

`make_plots.py` — **two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:**

- **QCD-like:** one hard collinear core (92% of p_T , $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8%, $\sigma = 0.30$). Each "spray" = n massive pions with exponential p_T fractions.
- **W-like:** two hard prongs with pair mass $m = \sqrt{z(1-z)} p_T$, $\Delta R = 80.4$ GeV ($z \in [0.30, 0.45]$), random orientation, + 6% soft contamination.

```
def _spray(rng, n, y0, phi0, pt_total, sigma):    # n pions around (y0,phi0)
    frac = rng.exponential(1., n); frac /= frac.sum()
    pt = pt_total * frac
    y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
    mt = np.sqrt(M_PI**2 + pt**2)
    return np.stack([pt*np.cos(phi), pt*np.sin(phi), mt*np.sinh(y), mt*np.cosh(y)], 1)
```

1500 events per sample. Purpose: QCD vs W is a *known* truth, so any observable that fails to separate them (or lands off the analytic curve) would be a real bug.

The toy parton shower (for the paper-figure closures)

`make_paper_plots.py` builds a **primary, fixed-coupling, leading-log** shower — emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta, \ln k_t)$ area, off a hard spine at $(y, \phi) = (0, \pi)$:

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}, \quad \text{keep } v \leq \ln(p_{T0}/2) - u \quad (z \leq \tfrac{1}{2}), \quad k_t > k_t^{\min}$$

This is **exactly the semi-classical picture the papers' analytic predictions are derived in** — which is what makes the closure *quantitative*, not just qualitative.

- **Jet areas:** one hard event + a dense grid of infinitely-soft **ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area (the anti-kt-paper construction).

$\bar{\alpha} = 0.25, p_{T0} = 1 \text{ TeV}, R = 0.4$, 20 000 showers. Everything is re-runnable with a fixed seed (`20260713`).

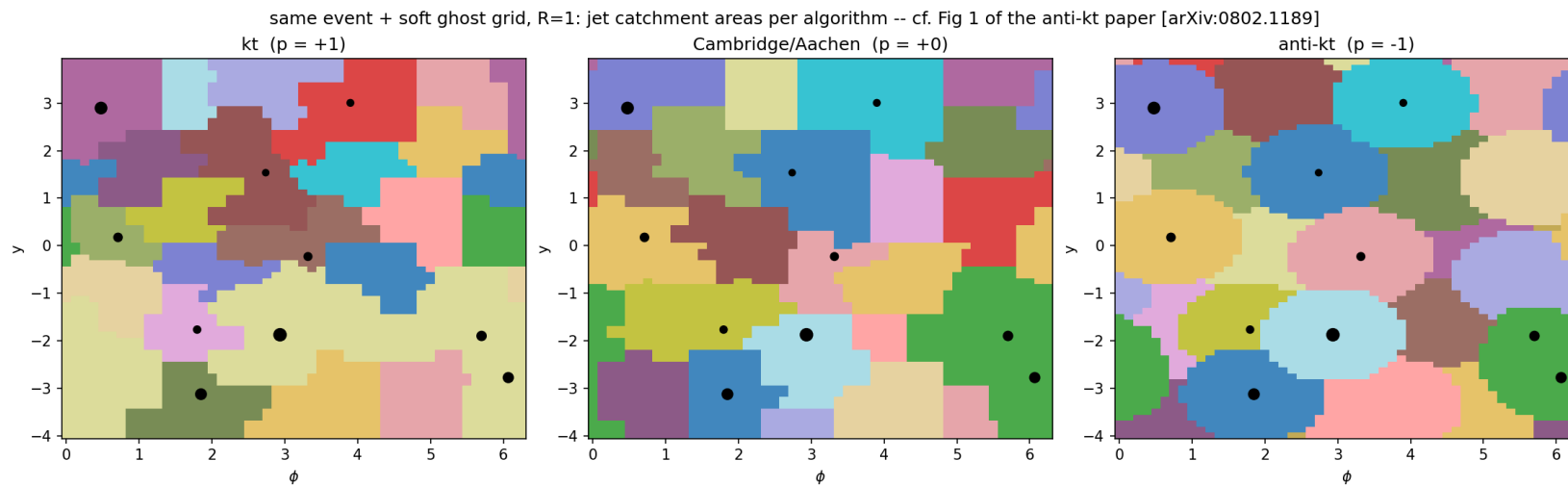
Correctness

each feature reproduces the signature figure of the paper it implements

Every slide states its **input**. Three input classes appear:

- (A) ad-hoc QCD/W toys (`make_plots.py`),
- (B) a leading-log toy shower (`make_paper_plots.py`),
- (C) real CMS PF-candidate constituents (`make_cms_plots.py`). All seed `20260713` .

anti-kt jet shapes — reproduces [0802.1189] Fig. 1



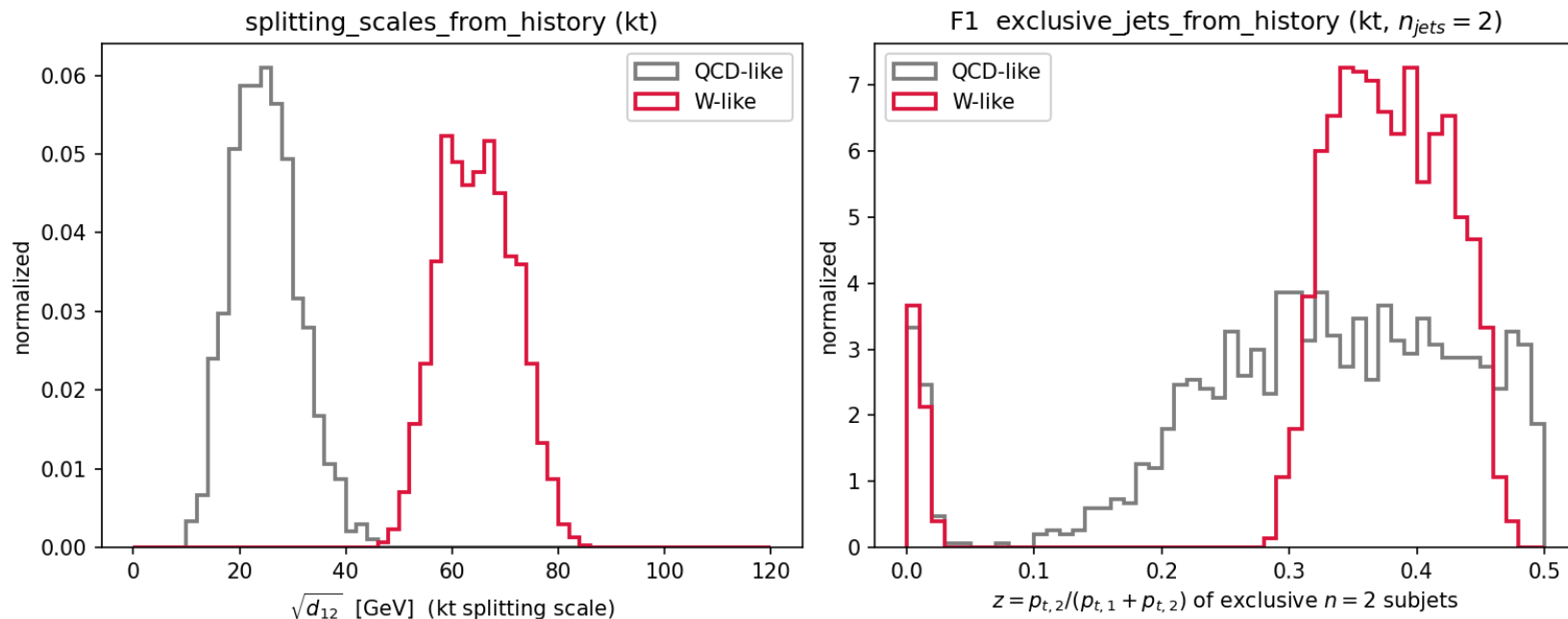
What: each algorithm's **jet catchment area**. kt & C/A give ragged, area-fluctuating jets; **anti-kt** gives rigid **circles** around each hard particle — the defining anti-kt result, from flashjet's own clustering.

How made — input (B), no substructure feature, clustering only.

1 synthetic event: **10 hard massless particles** (p_T 10–400 GeV, random y, ϕ) + a **uniform grid of ~3200 "ghosts"** ($p_T = 10^{-8}$, spacing 0.125 in y, ϕ). Clustered at $R = 1$ with `cluster_event_nn` for $p = +1/0/ - 1$; each ghost is coloured by the jet it lands in.

F1 — kt substructure separates 2-prong from QCD [0802.1189, 1111.6097]

kt-based substructure: hard two-prong structure separates from QCD [arXiv:0802.1189, 1111.6097]



Left — $\sqrt{d_{12}}$ (`splitting_scales_from_history`): kt scale of the last merge.

W-like peaks at $\sim m_W/2$, QCD low.

Right — **exclusive 2-subject z**

(`exclusive_jets_from_history` , $n_{jets} = 2$):

W-like balanced ($z \approx 0.35$), QCD lopsided ($z \rightarrow 0$).

How made — input (A), leading jet, kt clustering.

1500 **QCD-like** + 1500 **W-like** toy fat-jets ($p_T \approx 500$ GeV, $R = 0.8$); each a bag of

massive-pion 4-vectors from `_spray` — **synthetic particles, not real**

constituents. W pair-mass

= 80.4 GeV. Both observables read the **kt** merge history.

F2 — soft-drop z_g vs the analytic prediction [1402.2657]

The soft-drop momentum fraction z_g from

`groom_from_history`

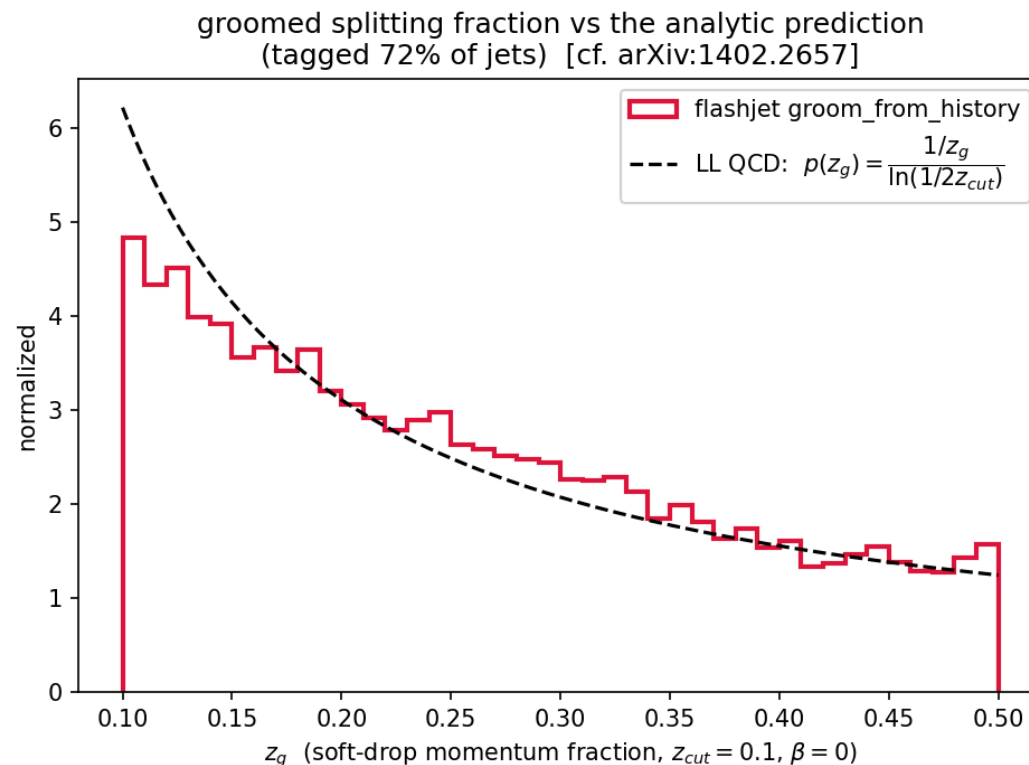
($z_{\text{cut}} = 0.1$, $\beta = 0$) tracks the **leading-log QCD prediction**

$$p(z_g) = \frac{1/z_g}{\ln(1/2z_{\text{cut}})}$$

across the **entire range**, no free parameters. 72% of jets tagged.

Decisive grooming-correctness plot: the target curve is *the* right answer, so lying on it is unambiguous closure.

How made — input (B). 20 000 toy-shower jets ($p_{T0} = 1$ TeV, $R = 0.4$), leading jet per event, soft-dropped by F2; the black curve is the analytic $1/z$ with *no* fit.



F2 — groomed mass ordering in β [1402.2657 Figs. 3-4]

Groomed mass $\rho = m^2/(p_T^2 R^2)$ for $\beta = 0, 1, 2$ vs ungroomed.

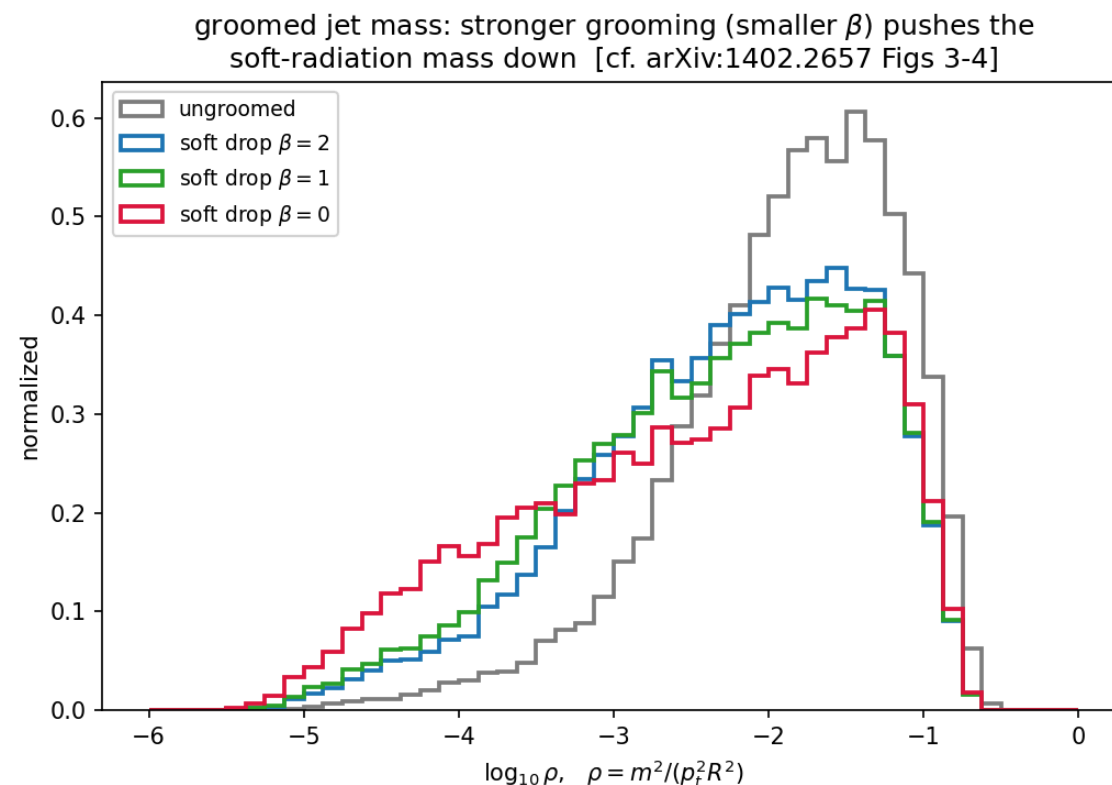
- Grooming shifts the spectrum to **lower** ρ (removes soft-wide radiation).
- **Smaller β grooms harder**: the $\beta = 0$ curve reaches furthest left.

Reproduces the β -ordering of the Soft Drop paper.

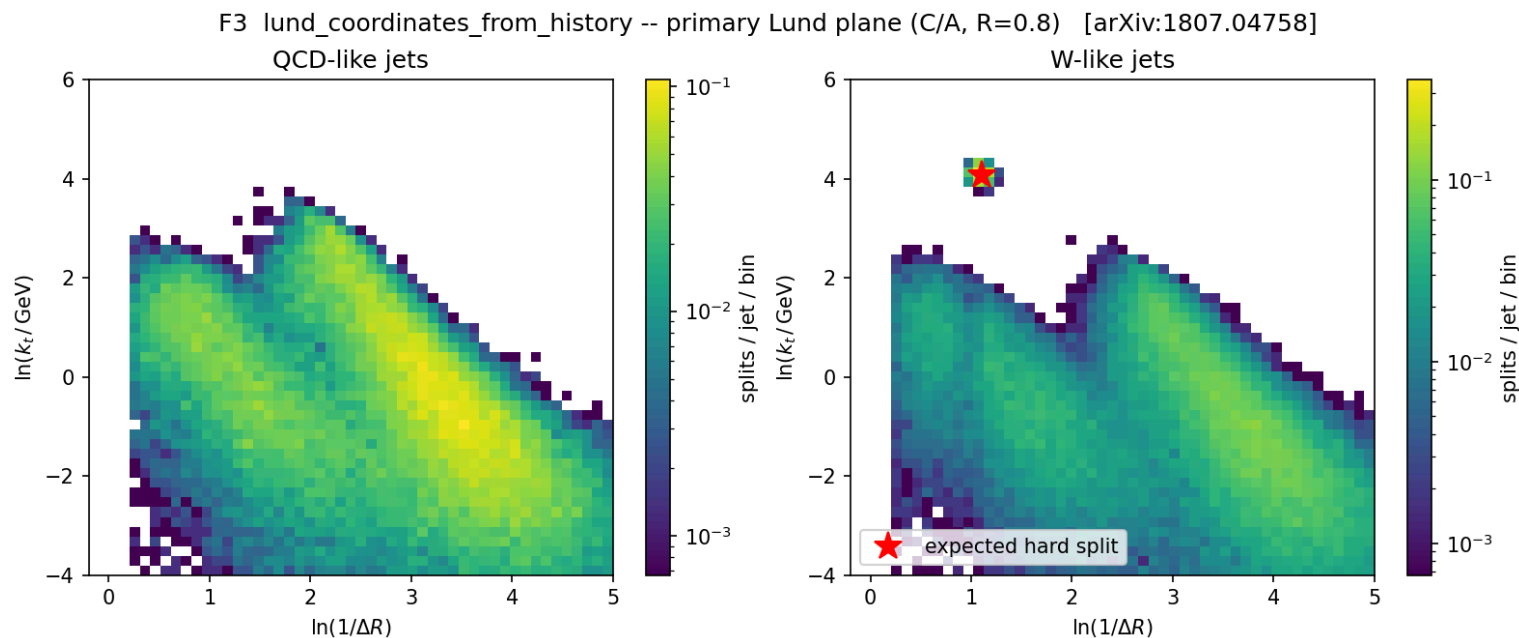
How made — input (B). Same 20 000 toy-shower jets; F2

`groom_from_history` re-run at

$\beta = 0, 1, 2$ (all $z_{\text{cut}} = 0.1$) on the leading jet, ρ from the groomed 4-vector.



F3 — primary Lund plane [1807.04758]



`lund_coordinates_from_history` (C/A , $R = 0.8$). **QCD** fills the soft-collinear region smoothly. **W-like** shows the same background **plus an isolated hard-splitting spot** exactly where the 2-body m_W decay must sit (red ★ = predicted position).

How made — input (A). Same 1500+1500 QCD/W toy fat-jets as the F1 slide, **C/A** clustered; F3 emits $(\ln 1/\Delta R, \ln k_t)$ for every primary split of the leading jet, histogrammed over all jets.

On real CMS data — input (C)

not toys any more: real detector-level PF-candidate constituents

The real-data pipeline (`make_cms_plots.py`)

Sample: `QCD_Pt-15to7000_Flat2018_pythia8` , **UL18 JMENano** (150X reprocessing) — the one NanoAOD flavour that stores the `PFCand` table + the `FatJetPFCand` constituent → AK8 map. Pulled via `DAS/ xrdcp` → `data/qcd_jmenano_150x.root` (1.7 GB). Ordinary NanoAOD has no constituents and is useless here.

These plots use *real jet constituents*, not toys:

1. Pick every CMS **AK8 FatJet** with $p_T > 300$, $|\eta| < 2.4$.
2. Gather its **PF candidates** via `FatJetPFCand_jetIdx→pfCandIdx→PFCand_{pt,η,φ,m}` .
`PFCand_pt/mass` are already **PUPPI-weighted** (the branch titles say so) — the weighted 4-vectors are stored directly, which is why no separate weight branch exists.
3. Feed *those constituents* to **our** `cluster(R=0.8, antikt)` — CMS AK8 is anti-kt $R = 0.8$.

4. Leading jet → **F2** soft drop ($z_{\text{cut}} = 0.1, \beta = 0$, declustering the **C/A** tree, as FastJet's SoftDrop does) — and **F3** Lund.
5. Chunked at 3000 jets (torch backend is $O(N^3)$; each AK8 jet is one independent event, so chunking is exact).

The stored CMS values are JEC-corrected: raw jet =

`FatJet_pt × (1 - rawFactor)` ;

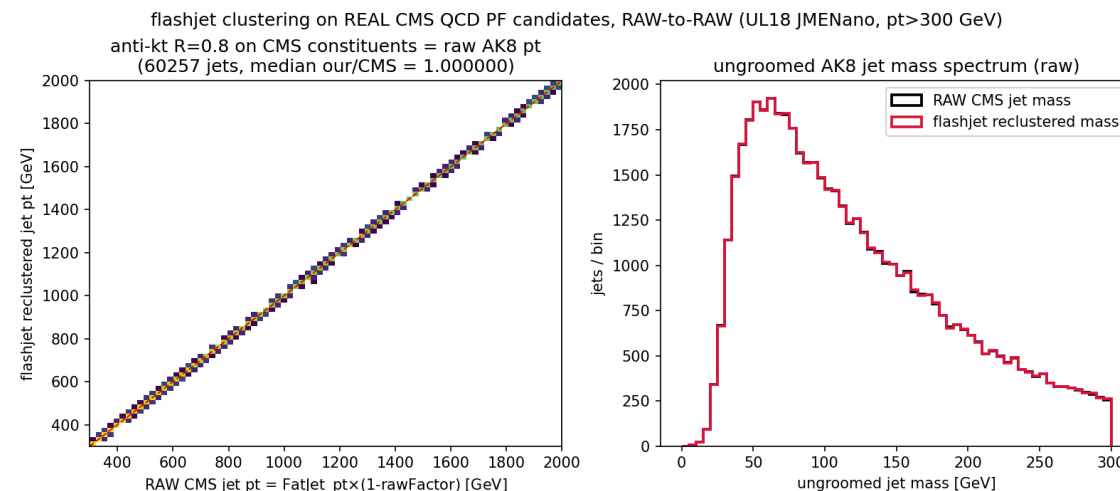
`FatJet_msoftdrop` = $m(\text{sub1} + \text{sub2})$ **with subjet JECs** (proven from data: $\Delta = +0.0002$ GeV).

CMS (1) — reclustering closes: our p_T vs CMS FatJet_pt

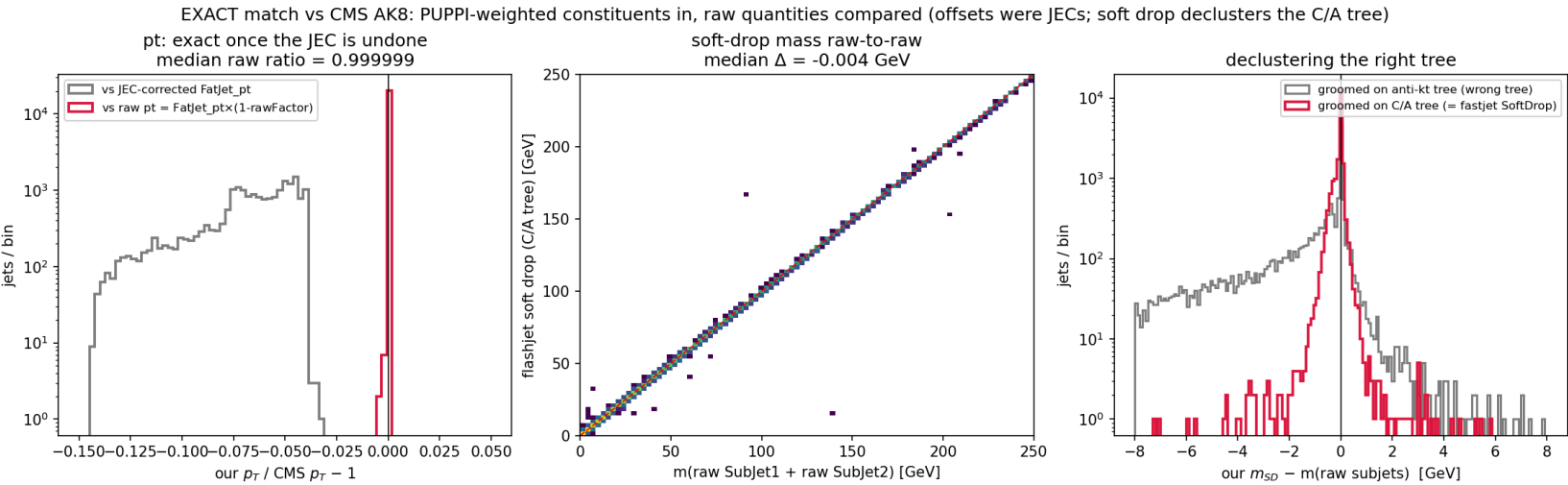
What: feed CMS's own AK8 constituents to **our** anti-kt $R = 0.8$ and compare the reclustered jet p_T to CMS's stored FatJet_pt, **jet-by-jet**.

Result — EXACT: vs the **raw** jet pt
(FatJet_pt $\times$ (1-rowFactor)) the ratio is
median 1.000000, $\sigma = 2.5\times 10^{-4}$ — pure NanoAOD storage precision. The apparent $\sim 6\%$ offset vs the stored value **is the L1L2L3 JEC**, nothing else: CMS stores corrected p_T , we recompute the raw one.

Input (C). 2D raw- p_T histogram + raw-mass spectra, real constituents (60257 jets);
run on HTCondor (cluster 9087059).



CMS (2) — soft-drop mass matches CMS EXACTLY (raw-to-raw)



The "offset" was never PUPPI — the constituents are *already* PUPPI-weighted. Two stacked artefacts, both removed: (1) JEC — `FatJet_pt / msoftdrop` are JEC-corrected; raw jet = `pt*(1-rowFactor)`, `msoftdrop` = $m(\text{JEC-corrected sub}_1 + \text{sub}_2)$, proven from data ($\Delta = +0.0002$ GeV).

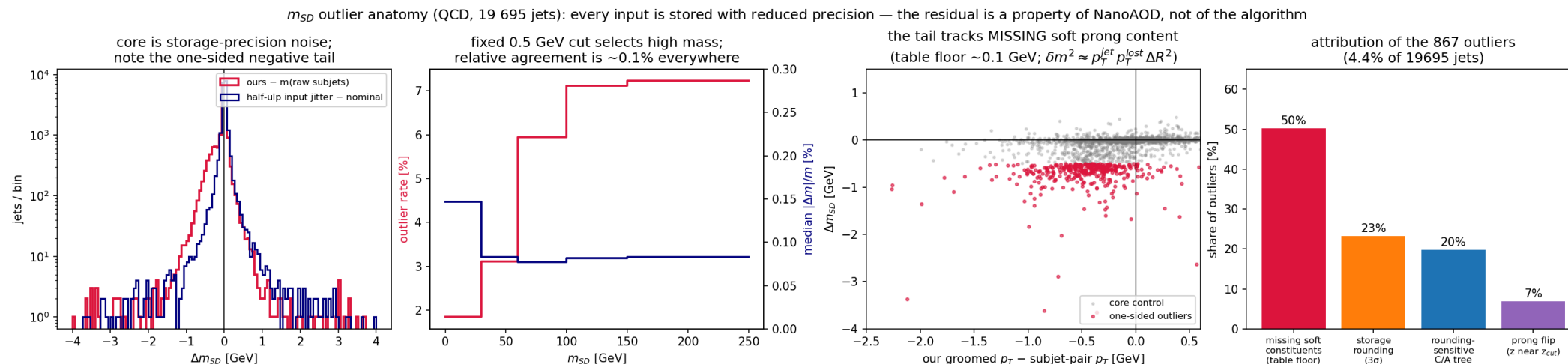
(2) Wrong tree — we groomed the anti-kt history; Fastjet SoftDrop declusters a C/A reclustering.

Raw-to-raw, C/A tree (20 065 jets):

	our – CMS
p_T ratio	0.999999 ($\sigma 2 \times 10^{-4}$)
m_{SD}	-0.004 GeV (95.6% <0.5 GeV)
z_g	$ \Delta = 7 \times 10^{-5}$

Residual = NanoAOD float precision. F2 reproduces CMS `msoftdrop`.

CMS (2b) — anatomy of the residual m_{SD} tail (the 4.4%)



Every input branch is stored with reduced mantissa (`PFCand_pt` ~ 10 bits $\approx 10^{-3}$,

`SubJet_pt/mass` ~ 9 bits, `SubJet_rawFactor` ~ 5 bits $\approx 2\%$) while CMS ran FastJet at full precision.

Attribution (per-jet tests, 19 695 jets): **50%** — soft candidates **missing from** `FatJetPFCand`

(the table has an effective ~ 0.1 GeV floor; the tail is one-sided, its groomed- p_T deficit correlates with Δm , and $\delta m^2 \approx p_T^{jet} p_T^{lost} \Delta R^2$ matches); **23%** within 3σ of

storage rounding; **20%** rounding-sensitive C/A trees (half-ulp jitter moves them); **7%** genuine

$z \approx z_{cut}$ prong flips (14 $\times$ enriched vs core). **Not fixable from NanoAOD — and not an**

algorithm error: the fixed 0.5 GeV cut just selects high-mass jets; relative agreement is $\sim 0.1\%$ everywhere.

Scripts: `outliers.py`, `analyze_outliers*.py`, `rerun_rest.py` (HTCondor 9098953).

CMS (3) — primary Lund plane of 60 257 real QCD jets

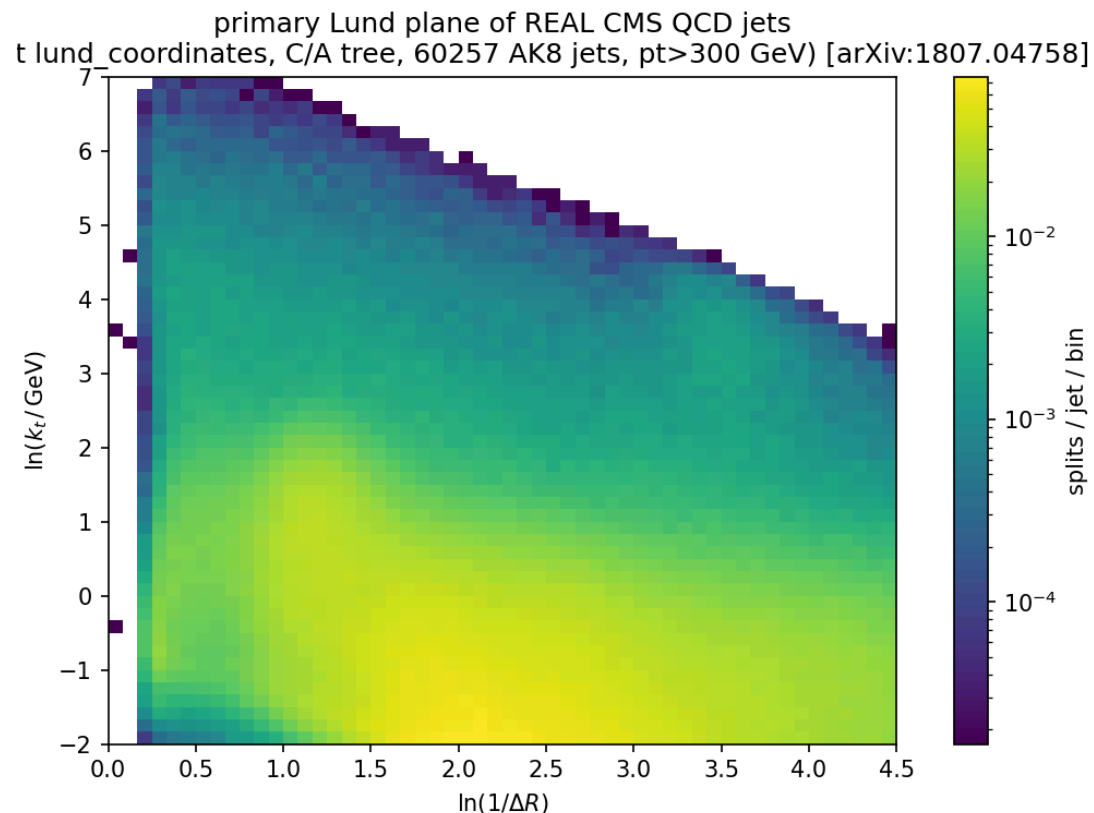
What: F3 `lund_coordinates` on the same real AK8 jets.

Needs no comparison curve — it *is* a clean, publication-quality primary Lund plane straight from detector-level simulation.

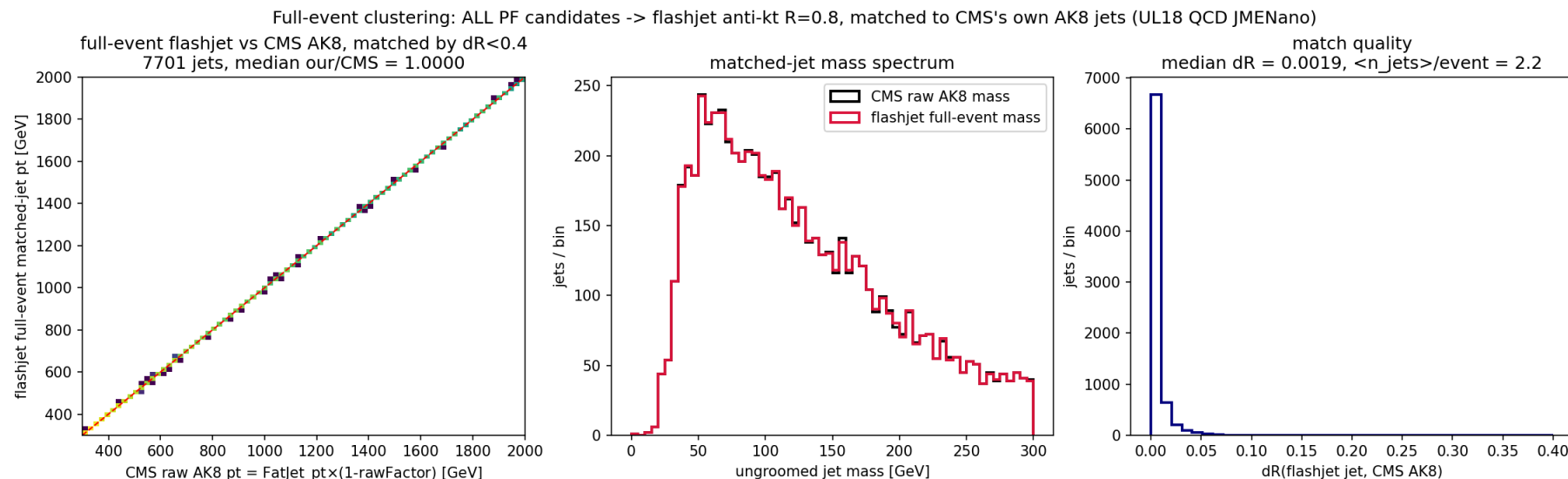
The full [1807.04758] structure emerges with **no toy**

input: the hard-collinear perturbative ridge, the soft plateau, and the three kinematic edges.

Input (C). 60 257 AK8 jets, $p_T > 300$ GeV, **C/A** reclustering; every primary split of each jet histogrammed.



CMS (4) — full-event clustering recovers CMS's own AK8 jets



What: the realistic path — feed **every** PF candidate in **the event** to flashjet anti-kt $R = 0.8$ (no `FatJetPFCand` per-jet pre-grouping), then match the resulting inclusive jets to CMS's **stored** AK8 jets (`FatJet_*` , which CMS clustered with FastJet) by $\Delta R < 0.4$.

Result (7 701 matched jets, $p_T > 300$, raw-to-raw): pt median 1.0000, 100% within 2%; match ΔR median 0.0019; mass spectra identical. Full-event flashjet reproduces CMS's own FastJet AK8 reconstruction to milliradian ΔR — no jets known a priori.

On a ttbar sample

a second, independent final state · comparison against the sample's own stored FastJet branches

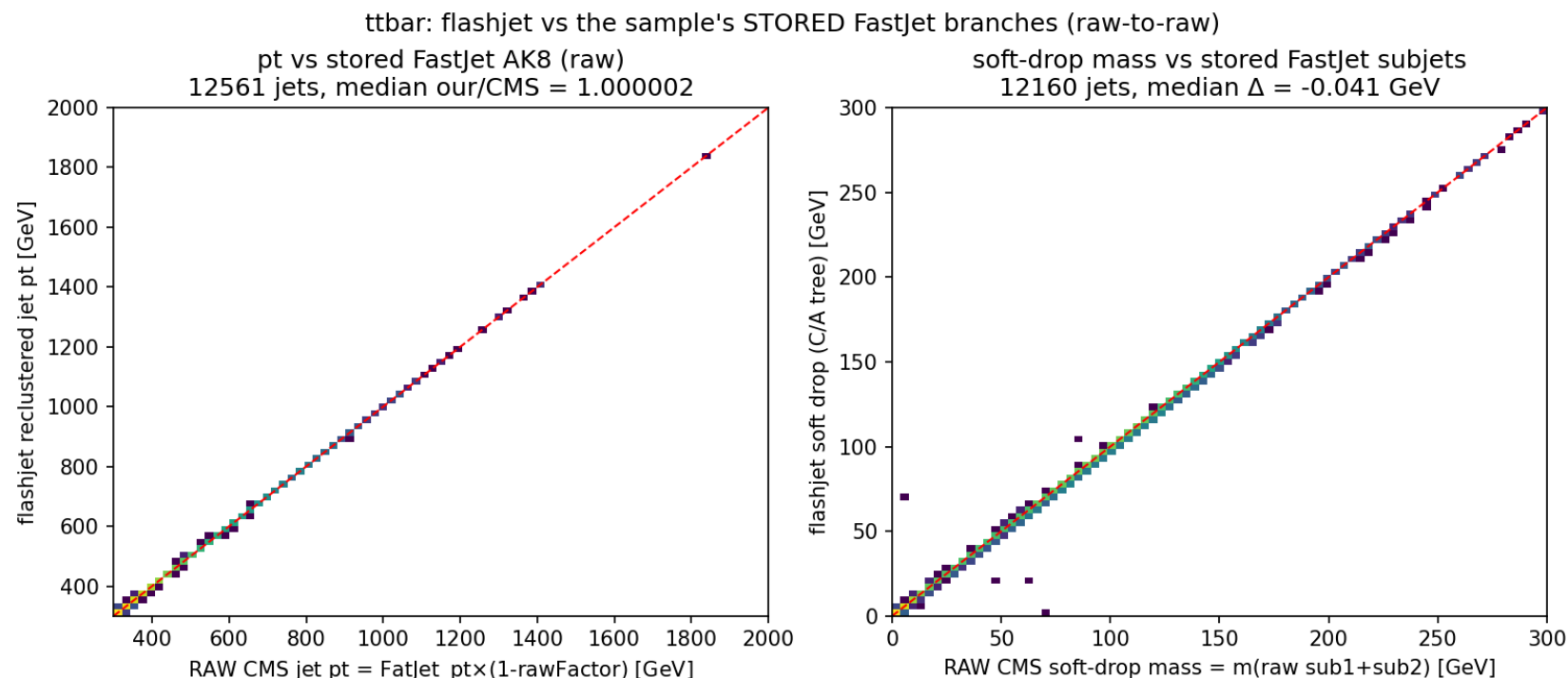
CMS TTTto2L2Nu UL18 JMENano 150X, 344 k events → 12 561 leading AK8 jets ($p_T > 300$, $|\eta| < 2.4$).

"Comparison against FastJet" = comparison against CMS's **stored** FatJet_* / SubJet_* / tau* branches

(CMS clustered them with FastJet), **not** a FastJet re-run.

Note: TTTto2L2Nu is **dileptonic** (both $W \rightarrow \ell\nu$), so these AK8 jets are mostly **b-jets + ISR**,
not hadronic top decays — a fully-hadronic sample follows in the next section.

ttbar — flashjet = stored FastJet branches (raw-to-raw)



pt (our anti-kt $R = 0.8$ vs `FatJet_pt*(1-rawFactor)`):
median **1.000002**, $\sigma 2.2 \times 10^{-4}$.

soft-drop mass (our C/A tree vs $m(\text{raw sub}_1 + \text{sub}_2)$):
median Δ **-0.041 GeV**, 94.2% < 0.5 GeV.

The exact match holds on a completely different final state (boosted tops + b-jets), not just QCD —
same conclusion, same NanoAOD-precision residual.

ttbar — primary Lund plane (F3)

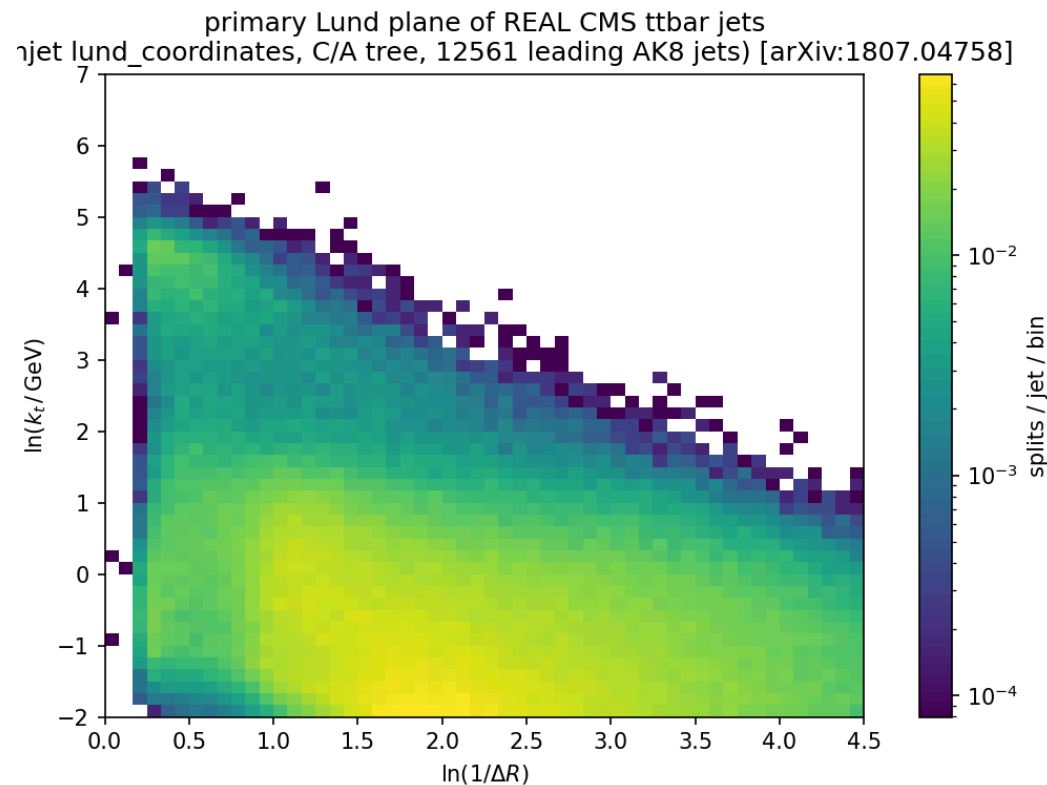
`lund_coordinates_from_history` (C/A, $R = 0.8$) on 12561 real ttbar AK8 jets.

The full [1807.04758] structure appears, and — unlike QCD — an **enhanced hard-splitting region** around $\ln 1/\Delta R \approx 1$, $\ln k_t \approx 1$. Since this sample is **dileptonic**

(no hadronic top/W decays), the enhancement comes from the different jet composition —

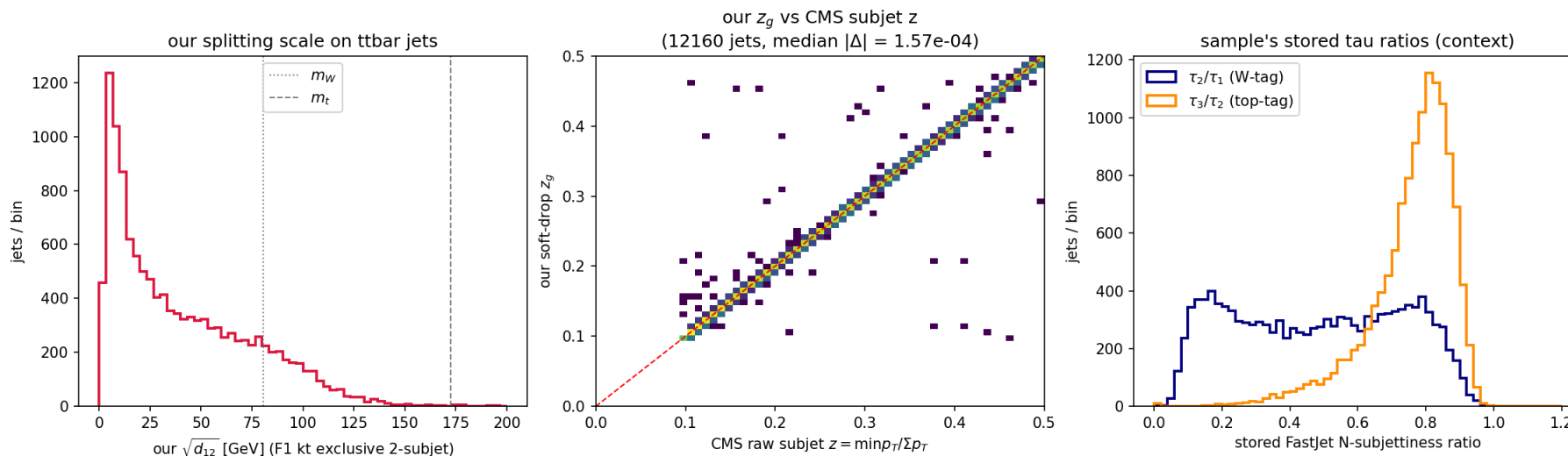
b-jets from the top decays + ISR — not from prongy top decays (those appear in the fully-hadronic sample, next section).

Clean, publication-quality Lund plane straight from detector-level ttbar simulation, entirely from our F3.



ttbar — our substructure vs stored FastJet observables

ttbar substructure: our F1/F2 observables + the sample's stored FastJet tau ratios



Left (F1): our exclusive-2-subjet $\sqrt{d_{12}}$ (kt splitting scale) on ttbar jets; the m_W/m_t lines are **mass-scale references only** — a decay splitting sits at $\sqrt{d_{12}} = m\sqrt{z/(1-z)} \approx m/2$, and this dileptonic sample has no hadronic decays.

Middle (F2): our soft-drop z_g vs CMS's **raw subjet** z , jet-by-jet — median $|\Delta| = 1.6 \times 10^{-4}$.

Right (context): the sample's stored FastJet **N-subjettiness** ratios — τ_2/τ_1 (W-tag), τ_3/τ_2 (top-tag). Our z_g tracking CMS's stored subjet z to 10^{-4} closes F2 directly against FastJet's SoftDrop output.

Three samples side by side

*QCD vs dileptonic $t\bar{t}$ bar vs **fully-hadronic** $t\bar{t}$ bar — the first sample with real boosted W /top decays*

(1) `QCD_Pt-15to7000_Flat2018` UL18 JMENano (13 TeV) — 160 393 jets ·

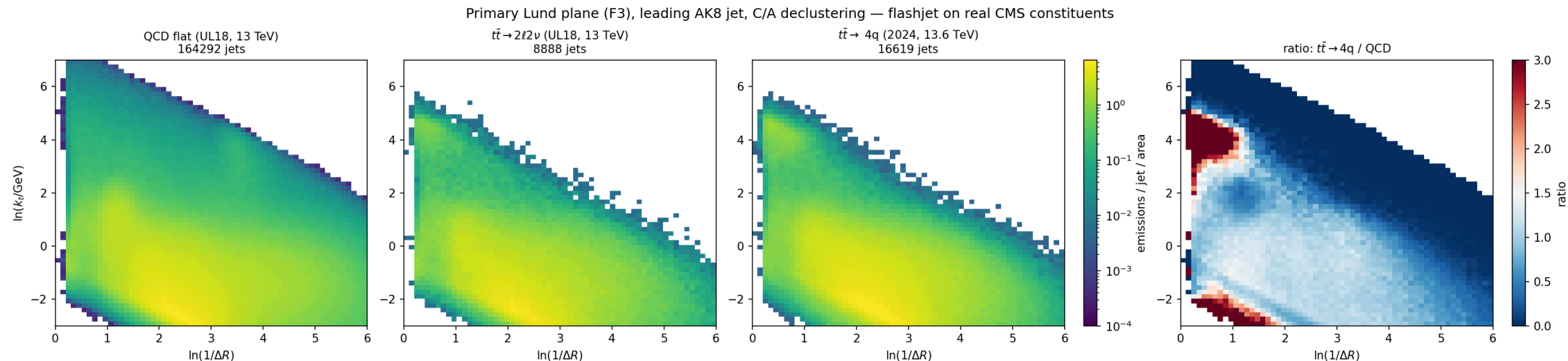
(2) `TTTo2L2Nu` UL18 JMENano (13 TeV) — 8 639 jets ·

(3) `TTto4Q` **RunIII2024Summer24 JMENanoV15** (13.6 TeV, 2024 pileup) — 16 516 jets.

All: leading AK8 jet/event, raw $p_T > 300$ GeV, $|\eta| < 2.4$, ≤ 200 constituents, full files

(`make_compare_plots.py` , HTCondor 9099026). Same schema everywhere: PUPPI-weighted `PFCand` + `FatJetPFCand` map.

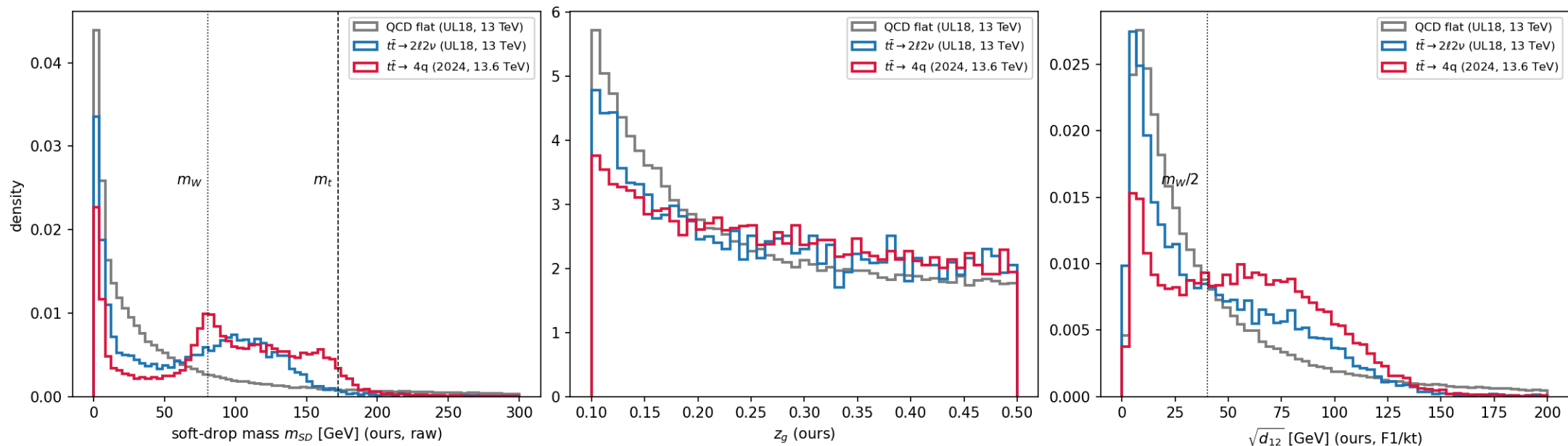
Lund planes: QCD → b-jets → boosted tops (F3)



The top-decay scale appears exactly where it must: the $t\bar{t} \rightarrow 4q$ plane grows a hard-splitting blob at $\ln k_t \approx \ln(m_W/2) \approx 3.7$, wide-angle ($\ln 1/\Delta R \lesssim 1$) — over **2×** QCD in the ratio panel (right). The dileptonic sample (b-jets, no hadronic top) shows only a mild version. Same clustering, same F3, three physics regimes.

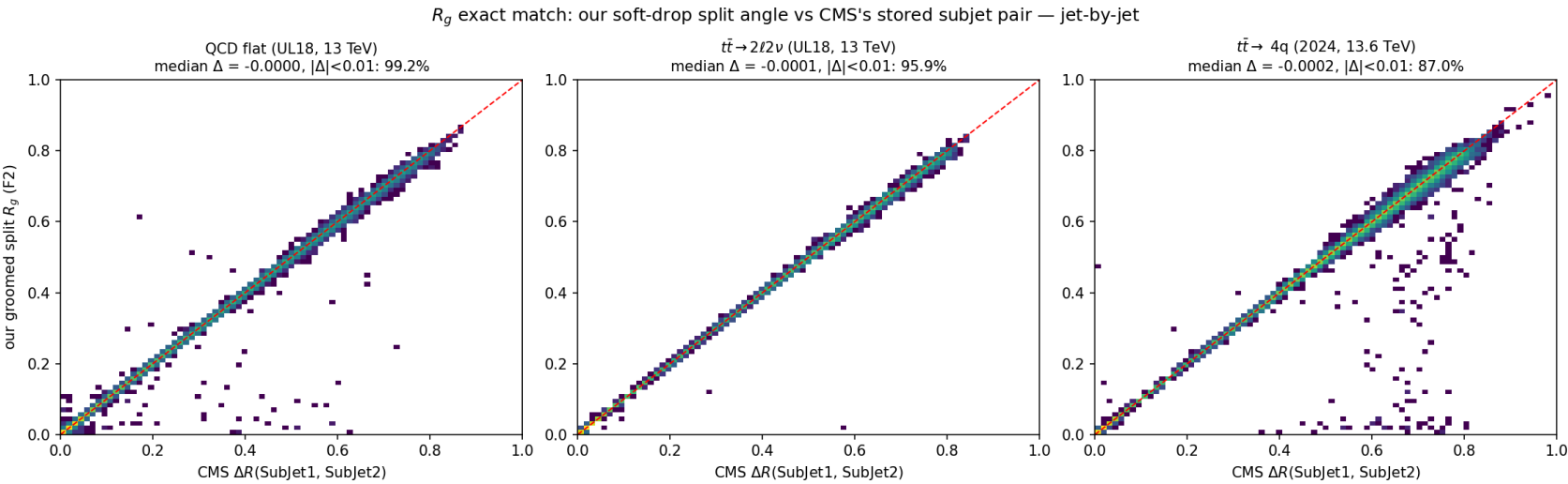
Spectra: m_W / m_t peaks from our grooming (F1 + F2)

Sample comparison, leading AK8 jet ($p_T^{raw} > 300$ GeV): QCD vs dileptonic vs fully-hadronic ttbar — all flashjet



Left (F2): our raw soft-drop mass — the fully-hadronic sample peaks at m_W with a top shoulder at m_t ; dileptonic peaks broad and lower (b + extra radiation); QCD falls. **Middle (F2):** z_g — ttbar flatter than QCD's $\sim 1/z$ (hard 2-body splits). **Right (F1):** kt splitting scale $\sqrt{d_{12}} = m\sqrt{z/(1-z)}$ — checked in m_{SD} slices: W-window jets peak at $39 \approx m_W/2$, top-window jets at $81 \approx m_t/2$ (which coincidentally $\approx m_W$!); QCD collapses to low scales.

R_g — a second jet-by-jet exact match (F2 vs stored subjets)



What: our split angle R_g (`groom_from_history` 's `dR`) vs stored $\Delta R(\text{sub}_1, \text{sub}_2)$, **jet-by-jet:** median $\Delta \leq 2 \times 10^{-4}$ — with z_g , both grooming observables close.

Run 3 = the outlier-anatomy mechanism at a higher dose: same floor/precision, but 2024 pileup →
90% of jets have p_T ratio < 1 (UL18: 50%), **99%** of Δm outliers negative;
relative agreement
stays **0.14%** — its jets are just heavy (median 93 GeV), so the 0.5 GeV window bites.

sample	p_T ratio	$m_{SD} \Delta$	$R_g \text{ } \Delta < 0.01$
QCD UL18	0.999999	-0.004 (95.7%<0.5)	99.2%
$t\bar{t} \text{ } 2\ell 2\nu$	0.999998	-0.033 (94.1%<0.5)	95.9%
$t\bar{t} \text{ } 4q \text{ '24}$	0.999556	-0.077 (69.3%<0.5)	87.0%

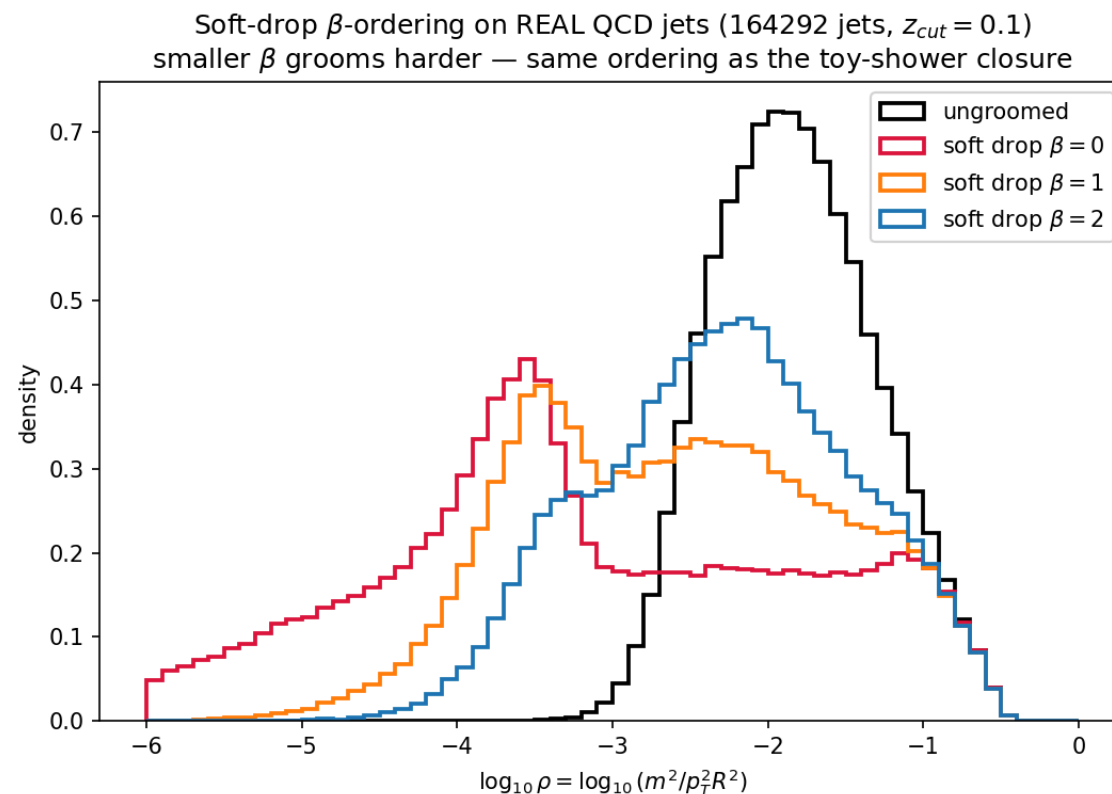
Soft-drop β -family on real QCD jets (F2)

The toy-shower closure, repeated on **164 292 real CMS jets**: re-groom the same C/A trees at $\beta = 0, 1, 2$ ($z_{\text{cut}} = 0.1$) and plot $\rho = m^2/(p_T^2 R^2)$.

- grooming pushes mass **down**; **smaller β grooms harder** — the exact ordering of Soft Drop [1402.2657] Figs. 3–4;
- $\beta = 0$ (mMDT) develops the characteristic flat low- ρ tail.

Input (C) — real constituents; grooming re-run 3× on the *same* merge histories

(a pure post-read: no re-clustering needed, which is the point of the history design).



Backup

Reproducibility & full-statistics numbers

All CMS plots regenerated **raw-to-raw on the C/A tree** at full statistics
(`make_cms_plots.py` , vectorized loader; run on **HTCondor** cluster 9087059, 60 257 jets):

observable	comparison	result
jet p_T (QCD)	our anti-kt vs <code>FatJet_pt*(1-rawFactor)</code>	median 1.000000 , σ 2.5×10^{-4}
soft-drop mass (QCD)	our C/A-tree vs $m(\text{raw sub}_1+\text{sub}_2)$	median -0.004 GeV , 95.7% <0.5 GeV
z_g (QCD)	our vs raw subjet z	$ \Delta = \mathbf{7\times 10^{-5}}$
jet p_T (ttbar 2L2Nu)	12 561 leading jets, HTCondor 9098883	median 1.000002 , σ 2.2×10^{-4}
soft-drop mass (ttbar)	our C/A-tree vs $m(\text{raw sub}_1+\text{sub}_2)$	median -0.041 GeV , 94.2% <0.5 GeV
full-event (QCD)	all PFCands $\rightarrow$ anti-kt, ΔR -match to <code>FatJet_*</code>	7 701 jets, pt 1.0000 , ΔR med 0.0019
R_g (3 samples)	our groomed <code>dR</code> vs $\Delta R(\text{sub}_1,\text{sub}_2)$, HTCondor 9099026	median $\Delta \leq 2\times 10^{-4}$; 99.2/95.9/87.0% <0.01
TTto4Q Run 3 2024	16 516 leading jets, raw-to-raw	pt 0.999556, m_{SD} -0.077 GeV — table-floor effect $\uparrow$ w/ pileup, rel. 0.14%

Residual = NanoAOD float storage throughout. Jets: anti-kt $R = 0.8$; grooming/Lund: big- R **C/A** reclustering (as FastJet).

Function reference (added)

```
exclusive_jets_from_history(hist_p1, hist_p2, hist_child, hist_d, mask,
                           n_jets=None, d_cut=None)           # F1
groom_from_history(hist_p1, hist_p2, hist_child, hist_d, mask, p4, R,
                   z_cut=0.1, beta=0.0, mu=None)             # F2 (soft-drop / mMDT / mass-drop)
lund_coordinates_from_history(hist_p1, hist_p2, hist_child, hist_d, mask, p4) # F3 -> (B,J,S,6)

# helpers
_resolve_roots(...)    # pointer-jump roots of an arbitrary sub-forest
_jet_roots(...)         # per-jet root pseudojet id, beam-merge order
_pseudojet_p4(...)      # E-scheme 4-momentum of every pseudojet, keyed by id
_dense_number_roots(...) # compact per-particle jet index
```

API surface: `ClusterOutput.exclusive_jets`, `.groomed_jets`, `.mass_drop`,
`.lund_coordinates`, plus a new `.mask` field to map slots → ids.

Reproducing everything

```
# on lxplus, env b_hive (torch 2.5.1)
cd /eos/home-c/cgupta/flashjet/plots/2026-07-13-substructure
micromamba run -n b_hive python make_plots.py          # F1/F2/F3 toy justification + parity
micromamba run -n b_hive python make_paper_plots.py    # anti-kt/SoftDrop/Lund paper figures
micromamba run -n b_hive python make_cms_plots.py      # real CMS UL18 QCD (needs qcd_jmenano_150x.root)
micromamba run -n b_hive python make_fullevent_plots.py # full-event clustering vs stored AK8
micromamba run -n b_hive python make_ttbar_plots.py    # ttbar (needs ttbar_jmenano_150x.root)
micromamba run -n b_hive python make_compare_plots.py  # QCD vs 2L2Nu vs TTto4Q comparisons
micromamba run -n b_hive python outliers.py            # m_SD outlier anatomy
# long jobs run on HTCondor: submit dir ~/flashjet_condor (AFS; /eos paths are
# rejected in submit files), executables cd to EOS + `micromamba run -n b_hive`

# tests
cd /eos/home-c/cgupta/flashjet/FlastJetDemo
micromamba run -n b_hive python -m pytest -q           # 85 passed, 13 skipped
```

Fixed seed 20260713 throughout. Papers catalogued in `References/Flashjet/papers.md`.